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Method of producing compost from animal waste

Abstract

A method of producing compost from animal waste may include placing animal waste in a waste pit. The method may include removing liquid from a top layer of the waste pit. The waste pit may include a bottom layer disposed at a bottom of the waste pit and adjacent the top layer. The bottom layer may be composed primarily of solids. The top layer may be composed primarily of liquid. After removing the liquid from the top layer of the waste pit, the method may include agitating the waste pit to mix the top layer and the bottom layer and create sludge. The method may include pumping the sludge into a system of channels in fluid communication with each other. Each of the channels may be sloped downwardly such that the sludge travels by gravity through the system of channels.

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Background/Summary

RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Patent Application No. 63/154,503, filed on Feb. 26, 2021, and entitled METHOD OF PRODUCING COMPOST FROM ANIMAL WASTE, which is incorporated herein in its entirety.

SUMMARY

(1) The present disclosure relates to a method of producing compost from animal waste, as well as related devices and systems. In some embodiments, a method of producing compost from animal waste may include placing animal waste in a waste pit. In some embodiments, the method may include removing liquid from a top layer of the waste pit. In some embodiments, the waste pit may include a bottom layer disposed at a bottom of the waste pit and adjacent to the top layer. In some embodiments, the bottom layer may be composed primarily of solids. In some embodiments, the top layer may be composed primarily of liquid. In some embodiments, after removing the liquid from the top layer of the waste pit, the method may include agitating the waste pit to mix the top layer and the bottom layer and create sludge.

(2) In some embodiments, the method may include pumping the sludge into a system of channels in fluid communication with each other. In some embodiments, each of the channels may be sloped downwardly such that the sludge travels by gravity through the system of channels. In some embodiments, the system of channels may be constructed of dirt or another suitable material.

(3) In some embodiments, the channels may be open-air. In some embodiments, each of the channels may be parallel to each other. In some embodiments, each pair of adjacent channels of the system of channels may be connected by a 180° bend. In some embodiments, the system of channels may include a start position at which the sludge enters the system of channels and an end position at which the sludge may stop flowing. In some embodiments, the system of channels may be sloped downwardly along an entirety of a pathway through the system of channels from the start position to the end position.

(4) In some embodiments, the start position may be disposed at or adjacent an end wall at an end of a first of the channels. In some embodiments, the end position may include another end wall, which may be disposed at an end of a last of the channels. In some embodiments, the channels may be separated by intermediate walls extending inwardly from an exterior wall. In some embodiments, the exterior wall may include the end wall at an end of the first of channels and/or the other end wall. In some embodiments, the exterior wall may be generally rectangular or square.

(5) In some embodiments, each of the channels may have a same downward slope. In some embodiments, each pair of adjacent channels of the system of channels may be connected by the 180° bend, and the 180° bend may be sloped downwardly. In some embodiments, each pair of adjacent channels of the system of channels may include a first channel in which the sludge flows in a first direction and a second channel in which sludge flows in a second direction opposite the first direction.

(6) In some embodiments, removing liquid from the top layer of the waste pit may include removing at least 80% of the liquid from the top layer. In some embodiments, removing liquid from the top layer of the waste pit may include removing between 80% and 90% of the liquid from the top layer. In some embodiments, the waste pit may be about 30 feet deep. In some embodiments, the top layer may be about 12 feet deep, and the bottom layer may be about 18 feet deep.

(7) In some embodiments, agitating the waste pit may include driving a motorized boat or similar vehicle in the waste pit. In some embodiments, the sludge may be pumped into the system of channels from the waste pit while the waste pit is agitated. In some embodiments, the method may include drying the sludge in the system of channels to form a compost between about 65% and about 70% moisture content level. In some embodiments, after drying the sludge in the system of channels to form the compost, the method may include removing the compost from the system of channels and arranging the compost in rows that are spaced apart. In some embodiments, after

removing the compost from the system of channels and arranging the compost in rows, the method may include using a compost tiller to turn the compost and/or remove moisture from the compost.
(8) It is to be understood that both the foregoing general description and the following detailed description are examples and explanatory and are not restrictive of the invention, as claimed. It should be understood that the various embodiments are not limited to the arrangements and instrumentality illustrated in the drawings. It should also be understood that the embodiments may be combined, or that other embodiments may be utilized and that structural changes, unless so claimed, may be made without departing from the scope of the various embodiments of the present invention. The following detailed description is, therefore, not to be taken in a limiting sense.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS
(1) Example embodiments will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:
(2) FIG. **1** is a schematic diagram of an example waste pit, according to some embodiments;
(3) FIG. **2** is a schematic diagram illustrating removal of liquid from a top layer of the waste pit, according to some embodiments;
(4) FIG. **3** is a schematic diagram illustrating agitation of the waste pit to mix the top layer and the bottom layer to create sludge, according to some embodiments;
(5) FIG. **4** is a schematic diagram illustrating pumping of the sludge into an example system of channels, according to some embodiments;
(6) FIG. **5** is a schematic diagram illustrating removal of compost from the system of channels and arranging of the composts in rows, according to some embodiments;
(7) FIG. **6** is a schematic diagram illustrating use of an example compost tiller to remove moisture from the compost, according to some embodiments;
(8) FIG. **7**A is an upper perspective view of an example system of channels;
(9) FIG. **7**B is a cross-sectional view along the line **7**B-**7**B of FIG. **7**A, according to some embodiments;
(10) FIG. **7**C is a cross-sectional view along the line **7**C-**7**C of FIG. **7**A, according to some embodiments; and
(11) FIG. **7**D is a cross-sectional view along the line **7**D-**7**D of FIG. **7**A, according to some embodiments.
DESCRIPTION OF EMBODIMENTS
(12) Referring now to FIG. **1**, in some embodiments, a method of producing compost from animal waste may include placing animal waste in a waste pit **10**, which may include a lagoon. In some embodiments, the animal waste may include organic matter, which may be obtained in farming. For example, the animal waste may include manure and/or urine from hogs or other animals. In some embodiments, other waste may be placed in the waste pit **10** with the animal waste. For example, vegetables, fruit, leaves, bark, or other items may be added to the waste pit **10**. In some embodiments, only plant and/or animal materials may be added to the waste pit **10**.
(13) Referring now to FIG. **2**, in some embodiments, the method may include removing liquid from a top layer **12** of the waste pit **10**. In some embodiments, the waste pit **10** may include a bottom layer **14** disposed at a bottom of the waste pit **10** and adjacent the top layer **12**. In some embodiments, the bottom layer **14** may be composed primarily of solids. In some embodiments, the top layer **12** may be composed primarily of liquid. In some embodiments, the top layer **12** and the bottom layer **14** may be formed after the animal waste is placed in the waste pit **10** and left for a period of time, allowing the solids in the animal waste settle to the bottom layer **14**.
(14) In some embodiments, removing liquid from the top layer **12** of the waste pit **10** may include

removing at least 80% of the liquid from the top layer **12**. In some embodiments, removing liquid from the top layer **12** of the waste pit **10** may include removing between 80% and 90% of the liquid from the top layer **12**. In some embodiments, the waste pit **10** may be about 30 feet deep or another suitable depth. In some embodiments, the top layer **12** may be about 12 feet deep, and the bottom layer **14** may be about 18 feet deep.

(15) Referring now to FIGS. **3-4**, in some embodiments, after removing the liquid from the top layer **12** of the waste pit **10**, the method may include agitating the waste pit **10** to mix the top layer **12** and the bottom layer **14** and create sludge **16**, illustrating in FIG. **4**, for example. In some embodiments, the sludge **16** may include a thick, soft, wet mud-like material or a semi-solid slurry. In some embodiments, agitating the waste pit **10** may include driving a motorized boat **18** in the waste pit **10**, which may facilitate efficient mixing of the top layer **12** and the bottom layer **14**. In some embodiments, the waste pit **10** may be agitated with another suitable machine or mechanism that mixes the top layer **12** and the bottom layer **14**.

(16) In some embodiments, the liquid from the top layer **12** may be removed by pumping the liquid out of the waste pit **10**. As illustrated in FIG. **4**, in some embodiments, the sludge **16** may be pumped into a system of channels **20** from the waste pit **10**. In some embodiments, the system of channels **20** may include multiple channels **22** in fluid communication with each other. In some embodiments, the liquid from the top layer **12** may be removed by pumping the liquid out of the waste pit **10** while the waste pit **10** is agitated, providing simultaneous sludge creation and transfer of the sludge **16** to the system of channels **20**.

(17) In some embodiments, the channels **22** may be open-air, which may facilitate drying of the sludge. In further detail, in some embodiments, a top of each of the channels **22** may be open, which may allow air and/or sunlight to dry the sludge **16**. In some embodiments, the system of channels **20** may be indoors or outdoors. In some embodiments, the channels **22** may also facilitate decomposition of the sludge **16** at a greater rate as the sludge **16** moves through the channels **22**.

(18) In some embodiments, the sludge **16** may be dried in the system of the channels **20** to form a compost that is between about 65% and about 70% moisture content level. In further detail, in some embodiments, the sludge **16** may be left in the system of channels **20** until the compost that is between about 65% and about 70% moisture content level is formed. In some embodiments, after drying the sludge **16** in the system of channels **20** to form the compost, the compost may be removed from the system of channels **20**.

(19) In some embodiments, a tractor **24** or another suitable machine may be used to remove the compost from the system of channels **20**. In some embodiments, the tractor **24** or the other suitable machine may arrange or pile the compost in rows **26**, which may facilitate further drying of the compost and/or access to the compost by a compost tiller. In some embodiments, a number of the rows **26** may correspond to a number of the channels **22** in the system of channels **20**, which may facilitate straightforward transfer of the compost by the tractor **24** from the system of channels **20** to an arrangement that promotes drying of the compost.

(20) Referring now to FIG. **6**, in some embodiments, the rows **26** may be parallel similar to the channels **22** and/or spaced apart to promote drying. In some embodiments, the compost tiller **28** may move down the rows **26** and/or in between the rows **26** to turn the compost, which may facilitate decomposition at a greater rate and/or remove moisture from the compost. In some embodiments, turning of the compost may activate the compost and enable the compost to heat up and break down more quickly.

(21) Referring now to FIGS. **7**A-**7**D, in some embodiments, the channels **22** of the system of channels **20** may be in fluid communication with each other. In some embodiments, each of the channels **22** may be sloped downwardly such that the sludge **16** (see, for example, FIG. **4**) travels by gravity through the system of channels **20**.

(22) In some embodiments, the system of channels **20** may include at least four of the channels **22**, which may provide for movement of the sludge **16** in a compact, efficient environment. In some

embodiments, the system of channels **20** may include four, five, or six of the channels **22**. In some embodiments, the system of channels **20** may include more than six of the channels **22**. In some embodiments, the system of channels **20** may include less than four of the channels **22**. As an example, FIG. **7**A illustrates a first channel **22***a*, a second channel **22***b*, a third channel **22***c*, a fourth channel **22***d*, a fifth channel **22***e*, and a sixth channel **22***f* (which may be referred to collectively in the present disclosure as "channels **22**").

(23) In some embodiments, the channels **22** may be progressively closer to the ground, upon which the channels **22** may sit. For example, the first channel **22***a* may be higher than the second channel **22***b*, which may be higher than the third channel **22***c*, etc. In some embodiments, each of the channels **22** may be parallel to each other. In some embodiments, each pair of adjacent channels of the system of channels **20** may be connected by a 180° bend **30**. For example, a particular 180° bend **30** may connect the first channel **22***a* and the second channel **22**, and another particular 180° bend **30** may connect the second channel **22** and the third channel **22***c*.

(24) In some embodiments, the system of channels **20** may include a start position **32** at which the sludge **16** enters the system of channels **20**. In some embodiments, the system of channels **20** may include an end position **34** at which the sludge **6** stops flowing. In some embodiments, a pump **33** may be proximate the start position **32** and may pump the sludge **16** from the waste pit **10** into the system of channels **20**. In some embodiments, the system of channels **20** may be sloped downwardly along an entirety of a pathway through the system of channels **20** from the start position **32** to the end position **34**. In these embodiments, the 180° bend **30** may be sloped downwardly, which may facilitate movement of the sludge **16** through the system of channels **20**. In other embodiments, the 180° bend **30** may be flat.

(25) In some embodiments, the start position **32** may be disposed at or adjacent an end wall **36** at an end of a first of the channels **22**. For example, as illustrated in FIG. **7**A, the start position **32** may be disposed at or adjacent an end of the first channel **22***a*. In some embodiments, the end position **34** may include another end wall **37**, which may be disposed at an end of a last of the channels **22**. For example, as illustrated in FIG. **7**A, the end position **34** may be disposed at an end of the sixth channel **22***f* In some embodiments, the channels **22** may be separated by intermediate walls **38** extending inwardly from an exterior wall **40**. In some embodiments, the exterior wall **40** may be generally rectangular or square, which may efficiently utilize space, or another suitable shape. In some embodiments, the exterior wall **40** may include the end wall **36** and/or the other end wall **37**. In some embodiments, the system of channels **20**, including the exterior wall **40** and/or the intermediate walls **38**, may be constructed of dirt or another suitable material. In some embodiments, the system of channels **20** may be constructed of metal, stone, plastic, or another suitable material.

(26) In some embodiments, each pair of adjacent channels of the channels **22** may include a first channel in which the sludge **16** flows in a first direction and a second channel in which the sludge **16** flows in a second direction opposite the first direction. For example, sludge **16** may flow the first direction in the first channel **22***a*, and in the second direction opposite the first direction in the second channel **22***b*.

(27) In some embodiments, some or all of the channels **22** may have a same downward slope, which may provide a consistent, steady flow of the sludge **16** for purposes of drying and decomposition. As an example, FIGS. **7**B-**7**D illustrate the first channel **22***a*, the second channel **22***b*, and the third channel **22***c* as having the same downward slope or incline. In some embodiments, downward slopes of the channels **22** may vary. In some embodiments, the downward slope of every other of the channels **22** may be in a same direction, and the downward slopes of the channels **22** in between the every other of the channels **22** may be in another same direction that is opposite the same direction.

(28) All examples and conditional language recited herein are intended for pedagogical objects to aid the reader in understanding the invention and the concepts contributed by the inventor to

furthering the art and are to be construed as being without limitation to such specifically recited examples and conditions. Although embodiments of the present inventions have been described in detail, it should be understood that the various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the invention.

Claims

1. A method of producing compost from animal waste, the method comprising: placing animal waste in a waste pit; removing at least 80% of liquid from a top layer of the waste pit, wherein the waste pit includes a bottom layer disposed at a bottom of the waste pit and adjacent the top layer, wherein the bottom layer is composed primarily of solids, wherein the top layer is composed primarily of liquid; after removing at least 80% of the liquid from the top layer of the waste pit, agitating the waste pit to mix the top layer and the bottom layer to create a sludge; and pumping the sludge into a system of channels in fluid communication with each other, wherein each of the channels is sloped downwardly such that the sludge travels by gravity through the system of channels.
2. The method of claim 1, wherein the channels are open-air.
3. The method of claim 1, wherein each of the channels are parallel to each other.
4. The method of claim 3, wherein each pair of adjacent channels of the system of channels are connected by a 180° bend.
5. The method of claim 1, wherein the system of channels comprises a start position at which the sludge enters the system of channels and an end position at which the sludge stops flowing, wherein the system of channels is sloped downwardly along an entirety of a pathway through the system of channels from the start position to the end position.
6. The method of claim 5, wherein each of the channels are parallel to each other, wherein each pair of adjacent channels of the system of channels are connected by a 180° bend, wherein the start position is disposed at or adjacent an end wall at an end of a first of the channels, wherein the end position comprises an end wall disposed at an end of a last of the channels.
7. The method of claim 1, wherein the channels are separated by intermediate walls extending inwardly from an exterior wall.
8. The method of claim 7, wherein the exterior wall is generally rectangular or square.
9. The method of claim 1, wherein each of the channels has a same downward slope.
10. The method of claim 1, wherein each pair of adjacent channels of the system of channels are connected by a 180° bend, wherein the 180° bend is sloped downwardly.
11. The method of claim 1, wherein each pair of adjacent channels of the system of channels comprises a first channel in which the sludge flows in a first direction and a second channel in which sludge flows in a second direction opposite the first direction.
12. The method of claim 1, wherein removing at least 80% of liquid from the top layer of the waste pit comprises removing between 80% and 90% of the liquid from the top layer.
13. The method of claim 1, wherein agitating the waste pit comprises driving a motorized boat in the waste pit.
14. The method of claim 1, wherein the sludge is pumped into the system of channels from the waste pit while the waste pit is agitated.
15. The method of claim 1, further comprising drying the sludge in the system of channels to form a compost between 65% and 70% moisture content level.
16. The method of claim 15, further comprising after drying the sludge in the system of channels to form the compost, removing the compost from the system of channels and arranging the compost in rows that are spaced apart.
17. The method of claim 16, after removing the compost from the system of channels and arranging the compost in rows, using a compost tiller to turn the compost.

18. A method of producing compost from animal waste, the method comprising: placing animal waste in a waste pit; removing liquid from a top layer of the waste pit, wherein the waste pit includes a bottom layer disposed at a bottom of the waste pit and adjacent the top layer, wherein the bottom layer is composed primarily of solids, wherein the top layer is composed primarily of liquid; after removing the liquid from the top layer of the waste pit, agitating the waste pit to mix the top layer and the bottom layer and create sludge; and pumping the sludge into a system of channels in fluid communication with each other, wherein each of the channels are parallel to each other, wherein each pair of adjacent channels of the system of channels are connected by a 180° bend, wherein each pair of adjacent channels and the 180° bend are sloped downwardly such that the sludge travels by gravity along an entirety of a pathway through the system of channels.
19. The method of claim 18, wherein the system of channels comprises a start position at which the sludge enters the system of channels and an end position, wherein the end position comprises an end wall disposed at an end of a last of the channels and configured to stop flow beyond the end wall.
20. The method of claim 19, wherein the end wall is constructed of dirt.